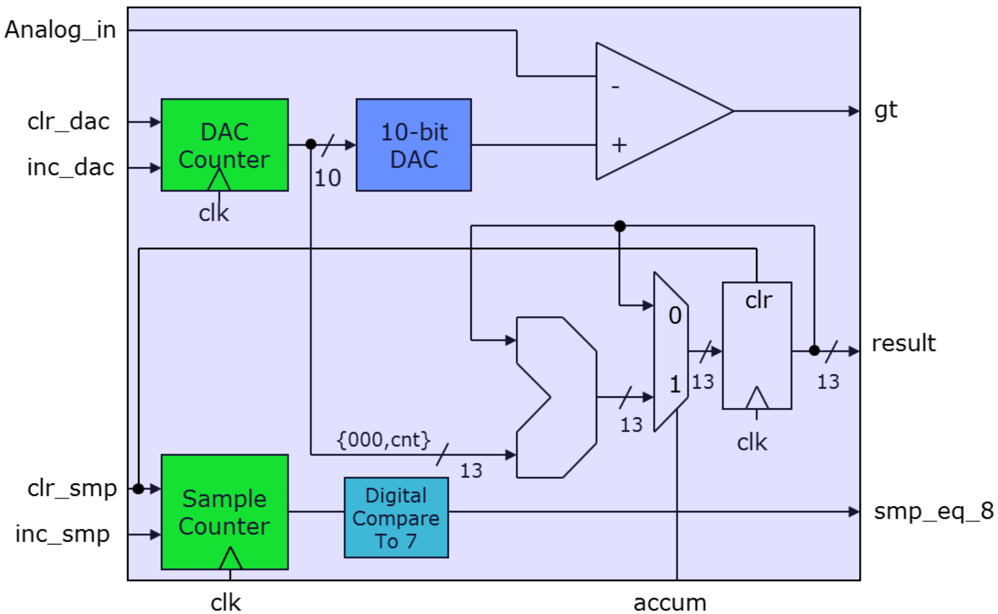
# ECE 551
## HW3

- Due Mon Mar 2nd @ 11:55PM
- Work Individually
- Remember What You Learned From the Cummings SNUG paper
- Use descriptive signal names and comment your code

# HW3 Problem 1 (15pts) SM Design

Not Started as Exercise…you are on your own for this one.

A Single Slope A2D converter (capable of averaging 8 samples) was discussed in Lecture1. The block diagram is shown below. On the course webpage under HW3 you will find files: **ss_A2D_tb.v**, **ss_A2D.v**, **ss_A2D_datapath.v**, **ss_A2D_SM_shell.sv**.




Flesh out **ss_A2D_SM_shell.sv** to complete the control (state machine). Simulate your resulting design using the provided self checking test bench. Submit your Verilog for the state machine (file should be called **ss_A2D_SM.sv**). Also submit proof that it worked.

Recall there is a video in week1 (SS_A2D) that discusses this design.

# HW3 Problem 2 (10pts) PWM12

Started as Ex09 on Mon Feb $16^{th}$

This problem has to do with the completion of **PWM12** used for motor drive.

See Exercise09 for complete specification.

Submit:
- **PWM12.sv**
- **Waveforms** showing you tested at various duty cycles. Make sure your username can be clearly seen in the images.

# HW3 Problem 3 (25pts) Dterm & full PID

Started as Ex10 on Weds Feb 18th

This problem has to do with the completion of the PID.

You need to infer the logic to implement the Dterm of the PID and test it. *(rudimentary self checking testbench).*

Then you need to integrate the Pterm (Ex06), Iterm(Ex08), and Dterm together to form the full PID.

See Exercise10 for complete specification.

Submit:

- **Dterm.sv** & **Dterm_tb.sv** *(stand alone Dterm logic and testbench)*
- **PID.sv** & **proof** *(transcript window)* it passed the provided testbench. Make sure your username can be clearly seen on the image.

# HW3 Problem 4 (**20pts)** UART Transmitter

Started as Exercise11 on Mon Feb 23rd

Implement a the UART Transmitter (**UART_tx.sv**).

Make a simple test bench for it. This is one instance in which I would not spend too much time on the test bench. You can just instantiate your transmitter and send a few bytes. Verify the correct functionality (including baud rate) by staring at the green waveforms. You will make a more comprehensive test bench in the next problem.

Submit **UART_tx.sv**

# HW3 Problem 5 (**20pts + 10pts)** UART Receiver

Started as Exercise12 on Wed Feb 25th

Implement a the UART Receiver (**UART_rx.sv**).

Since you have a transmitter too, it is now easy to make a self checking test bench. Architect the test bench as shown. Is does the 8-bit value you transmit match the value you receive when the transmission completes.

Submit **UART_rx.sv** and your self-checking test bench (and **proof** it ran). Make sure your username can be clearly seen in the images.